

Tracer: An Integrated Autonomous Sensing System for Pre-Dig Subsurface Utility Mapping

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1 Introduction

Tracer is an autonomous ground rover that maps buried utilities before excavation begins. It combines ground-coupled ground-penetrating radar (GPR), electromagnetic induction (EMI) sensing, RTK-GNSS positioning, and LiDAR-based localization on a single mobile platform, and processes the fused sensor data through an onboard interpretation pipeline that outputs a georeferenced, confidence-scored utility map. The system is designed for the workflow gap between record-based utility information (ASCE 38 Quality Levels D and C) and physically exposed verification (Quality Level A): the geophysical designation step, QL-B, which today depends on technicians manually walking survey carts across a site. Tracer automates the collection and first-pass interpretation of that data and delivers CAD/GIS-ready outputs, so that subsurface utility engineering (SUE) firms and contractors can make excavation decisions from consistent, spatially registered evidence rather than incomplete records.

The stakes of this problem are well documented. The Common Ground Alliance recorded 196,977 reported damages to buried infrastructure in a single year, with total direct and indirect economic damage estimated between \$30 billion and \$61 billion annually in the United States [1]. Contractor-side data shows the failure is not only one of awareness: 64% of reported utility strikes occurred where the line was at least two feet outside the marked area, and 87% of contractors reported waiting more than one business day for locators to arrive [2]. The bottleneck is the manual, labor-constrained process by which subsurface geophysical data is collected and interpreted.

In this whitepaper, we describe the Tracer system in full. Section 2 provides background on the subsurface utility problem and the quality-level framework that structures the industry; Section 3 describes the system architecture and its five subsystems; Section 4 details the core technical contribution, the four-stage subsurface interpretation pipeline; Section 5 presents evaluation results; and Sections 6–8 cover external validation, limitations, and the roadmap. Tracer is a working prototype, developed with Eratos Robotics and evaluated on controlled test sites with buried reference targets. It is not yet certified for QL-B deliverables under ASCE 38-22, and results should be read as engineering evaluations rather than field validation.

2 Background: The Subsurface Utility Problem

Every excavation begins with the same question: what is under the ground here? The existing answer is fragmented and insufficient. Utility records are frequently incomplete, outdated, or missing depth and geometry entirely. Public one-call (811) systems mark public utilities, but coverage stops at the meter or right-of-way, response is slow, and marks are approximate. Private utilities — the internal networks of data centers, hospitals, campuses, and industrial sites — receive no public locating support, and the contractor bears direct liability for striking them.

The industry formalizes this uncertainty through the ASCE 38 quality-level framework [3, 4]. QL-D is information from existing records; QL-C adds correlation with surface-visible features; QL-B is designation

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by surface geophysical methods, primarily GPR and EMI; QL-A is precise location by physical exposure. The structural bottleneck sits at QL-B: it is the level at which excavation decisions become defensible, but producing it requires trained technicians to manually push sensor carts across a site in a grid and then manually interpret the resulting radargrams. Both steps are slow, labor-constrained, and operator-dependent — a small tilt in a handheld antenna can shift an apparent utility position by roughly a foot. The most decision-critical layer of subsurface information is therefore also the hardest to produce at scale.

Tracer addresses this bottleneck directly: standardize QL-B-grade data collection with an autonomous platform, and reduce the interpretation burden with an automated first-pass analysis that human experts review rather than produce from scratch.

3 System Architecture

Tracer is organized as five subsystems on a single rover platform, connected by a common timing and georeferencing backbone (Figure 1).

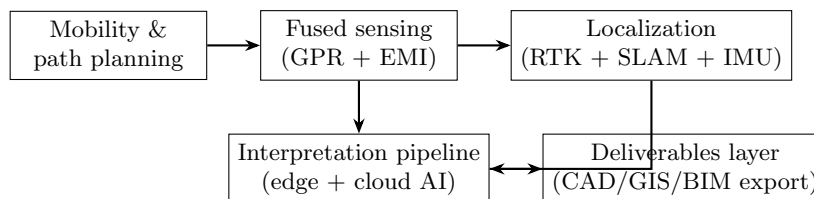


Figure 1: Tracer system architecture: coverage-optimal scanning, time-synchronized sensing and localization, automated interpretation, and engineer-ready export.

3.1 Mobility and path planning

The rover chassis is a ruggedized ground platform sized to carry the full sensor payload while keeping the GPR antenna close to the ground at a stable height, since ground coupling and mechanical stability directly determine data quality. A coverage planner generates a lawnmower-pattern route over an operator-defined boundary, and the rover follows it at constant velocity, giving uniform trace density that a human pushing a cart cannot maintain.

3.2 Fused sensing stack

Two complementary geophysical sensors run continuously during a scan. Ground-coupled GPR detects dielectric contrasts and is the primary means of finding nonmetallic utilities such as PVC conduit; an active EMI sensor detects conductive targets and traceable lines that GPR alone can miss. The modalities cover each other’s blind spots: GPR degrades in highly conductive soils where EMI still performs, and EMI is blind to nonmetallic targets that GPR can resolve. The EMI unit is mechanically isolated from the chassis to limit interference from the rover’s own electronics.

3.3 Localization

Every sensor trace is stamped with a position from a tightly coupled fusion of RTK-GNSS, LiDAR SLAM, IMU, and wheel odometry. RTK provides centimeter-grade global coordinates in open sky; SLAM and odometry maintain positioning through GNSS-degraded zones near structures. This turns a stream of radargrams into a spatially registered dataset rather than a strip chart.

3.4 Interpretation pipeline

The interpretation pipeline (Section 4) converts synchronized GPR and EMI returns into detected utility segments with confidence scores. First-pass inference runs on the rover’s edge computer (NVIDIA Jetson class) during the scan; deeper processing and final classification run in the cloud after upload.

3.5 Deliverables layer

The output layer packages detections into products a project team can act on: a utility path map color-coded by type, per-segment confidence scores, flagged ambiguous zones, recommended potholing locations, and exports to CAD, GIS, and BIM formats used in existing workflows. A human SUE analyst reviews and approves the outputs; Tracer reduces the interpretation burden rather than removing the expert.

4 The Subsurface Interpretation Pipeline

The core of the platform is the pipeline that turns raw geophysical returns into a confidence-scored utility map. Manual radargram interpretation is the scarcest skill in the current workflow, and automating its first pass is where most of the system’s value concentrates. It runs in four stages.

1. **Preprocessing and clutter removal.** Raw GPR B-scans are time-zero corrected, gain-normalized, and background-subtracted. Frequency–wavenumber (F–K) filtering strips coherent soil clutter, isolating the hyperbolic reflection signatures of discrete buried objects. EMI channels are drift-corrected against a per-site baseline.
2. **Migration.** A back-projection migration collapses each reflection hyperbola into a compact response at the true target location, using a soil velocity estimate fit from the hyperbola geometry itself, converting travel-time data into depth-registered imagery.
3. **Detection and classification.** A convolutional neural network runs over the migrated imagery to detect candidate utilities, estimate depth range, and classify probable material (metallic vs. nonmetallic) using the paired EMI response as an additional input channel. Each detection carries a calibrated confidence score rather than a binary label.
4. **Georeferenced fusion and mapping.** Detections from overlapping scan lanes are associated and fused in the site coordinate frame using the localization stream, linking them into continuous utility segments with aggregate confidence. Zones where evidence conflicts or coverage is thin are flagged as ambiguous and surfaced as potholing recommendations.

The stage boundaries are deliberate: stages 1–2 are physics-driven, stage 3 is learned, and stage 4 is geometric, which keeps the learned component small and auditable during review.

5 Evaluation

All results come from controlled test-bed experiments on outdoor plots with buried reference targets at surveyed positions (Q1–Q3 2026). They are engineering evaluations of the prototype, not field validation on active construction sites.

Table 1: Prototype evaluation results on controlled test beds.

Metric	Result	Measurement method
Horizontal positioning accuracy	± 5 cm RMS	Detections vs. surveyed target positions
Metallic target detection rate	91%	Buried metal pipe/conduit, depths to 1 m
Nonmetallic (PVC) detection rate	78%	2-inch PVC in sandy loam, depths to 0.6 m
False positive rate	9% per scan lane	Detections with no corresponding target
Path-following cross-track error	< 10 cm	Planned vs. executed scan route (RTK log)
Collection speed vs. manual baseline	$\approx 5\times$	Area per hour vs. single-channel cart walk

The gap between metallic (91%) and nonmetallic (78%) detection reflects the physics of the problem — small PVC targets produce weak reflections — and closing it in high-attenuation soils is the central technical risk of the program. The internal target for a commercially defensible system is above 95% true positive and below 5% false positive on nonmetallic targets in difficult soils; the prototype is not there yet, and we state this plainly.

6 External Validation

Eratos Robotics is backed by 1517 Fund and Emergent, and the team built the current prototype through the Founders, Inc. Canopy program. The company is in active letter-of-intent discussions with SUE firms and contractors for outdoor field pilots on closed private sites, where liability for private utilities is concentrated. No pilot revenue has been recognized to date, and we do not claim commercial traction beyond these engagements.

7 Limitations

We note the following limitations candidly.

1. **Test-bed scope.** All detection results come from controlled plots with known targets in favorable soils; performance on active sites with debris, fill, and utility congestion is unmeasured.
2. **Nonmetallic detection in difficult soils.** Detection of small PVC targets in wet clay and other high-attenuation soils is unproven and is the program’s primary technical risk.
3. **Semi-autonomy.** An operator must define scan boundaries, supervise operation, and review outputs; claims of unattended operation would be premature.
4. **No certification or liability transfer.** Outputs are not PE-stamped, carry no errors-and-omissions coverage, and do not yet constitute certified QL-B data under ASCE 38-22; they are decision-support intelligence, not a legal substitute for professional locating.
5. **Hardware endurance.** Mean time between interventions on rough terrain has not been characterized over long deployments.

8 Roadmap

The near-term plan follows fixed milestones. Q3 2026: outdoor field pilots with SUE partners on live private sites, validating detections against potholed ground truth and establishing a professional-engineer review workflow. Q4 2026: paid pilot contracts, E&O coverage, and production CAD/BIM/GIS export formats. 2027: fleet deployment under a robotics-as-a-service model, with each deployment adding paired radargram-and-ground-truth data that improves the interpretation models; the long-term advantage is this dataset and interpretation layer, not the rover chassis.

The architecture ports to adjacent domains largely unchanged: recurring facility digital twins and asset handoff for campus owners, excavation risk planning software for contractors, insurer-facing risk reporting, and broader subsurface infrastructure inspection.

9 Conclusion

Tracer began as a first-principles investigation into why utility strikes persist despite decades of one-call systems, and it landed on a specific answer: QL-B geophysical data is bottlenecked by manual collection and manual interpretation. Tracer automates both while keeping experts in the loop for judgment and review. The prototype works on controlled ground; the next milestones test whether it holds up in the field.

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